Time or Money? Togetherness and Intrahousehold Allocation *

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Abstract

Joint experience between partners, or togetherness, are key benefits of marriage. Despite the importance, the allocation tradeoffs between togetherness and other household activities, and its relation to gender differences, remains limited understood. This paper captures time use and consumption allocation between spouses, with a particular focus on togetherness. The study shows that women place higher value on home-production, while men hold more bargaining power. This study also reveals a substitutable role between joint leisure time and expenditure. These patterns help explain the gender gap in time use and consumption within families.

Keywords: togetherness, gender inequality, intrahousehold allocation, LISS data, consumption, leisure.

JEL classifications: D13, D79, J1.

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1 Introduction

This paper studies the intrahousehold allocation of time and expenditure, with a specific focus on the togetherness value of married couples. Togetherness takes the form of joint experiences between partners and is one of the major gains of marriage. The prevailing family economics emphasize the role of marriage from material perspectives, e.g. household specialization, economies of scale, and risk sharing, but often overlook the emotional perspective, e.g. the desire to be together with a spouse. Given the significance togetherness plays in marriage stability and children's development, this paper investigates the value spouses attach to their joint experiences and the way households allocate time and expenditure for their togetherness.

The main innovation of this paper is to develop an empirically tractable model that accounts for time and money investment in togetherness, and, in particular, to understand whether time and money are substitutes or complements in producing the togetherness value between spouses. Prior literature overlooks the expenditure, but solely concentrates on joint leisure time that spouses physically together. This naturally leads to the absence of a monetary dimension in togetherness evaluation and thus inaccurate analysis of intrahousehold behaviours, e.g., underestimated marriage gain and overstated working schedule synchronization. Furthermore, the discussion about time and money substitutability/complementarity contributes to our understanding of marriage stability and relationship dynamics. If money and time are substitutes for providing togetherness value (pleasant joint experiences), spouses can easily replace companion time with more luxurious services e.g., luxurious hotels or expensive restaurants, without harming their relationship. This argument indicates spouses are sensitive to wage increases and it provides significant implications for labour policy. If money and time are complements, spouses should invest much joint time and expenditure to maintain the joy of togetherness. We then find that money cannot replace time, which provides supporting evidence for time synchronization and flexible working schedules of spouses.

This study also enhances our understanding of spouses' gender-related preferences and joint decision-making processes within families. This paper develops a parametric collective model to study how couples, with different preferences, jointly allocate their time and money across private use, market work and domestic productions: childcare, chores and spouses' togetherness, at the household level. There are two channels in the intrahousehold allocation, couples' diverging preferences and their bargaining power within the family. The estimation result reveals which mechanisms and which part of preference heterogeneity leads to the gender differences in behaviours, in particular behavior patterns in the traditional family.

Overall, this study is important in two aspects. First, it provides implications for marriage satisfaction by drawing attention to couples' togetherness quality- proxy as money spent on the togetherness activities, in addition to the quantity- joint leisure time spent together. Second, it presents the mechanism and main sources of unequal time and money division between couples, including the husbands' and wives' differences in market work participation and enrollment in domestic unpaid work.

Model. We start from a collective model with household production in the spirit of Cherchye et al. (2012). An appealing feature of the collective model is that it recognizes the household consists of multiple individuals who each have their own rational preferences. These individuals are assumed to take Pareto efficient decisions resulting from an intrahousehold bargaining process. Each household has three types of home production: children's welfare, spousal togetherness and chores, which are produced by means of expenditures and time invested in the corresponding activities. The individuals in the household value their own private leisure, private consumption, and the outputs of the three productions. The household then decides the allocation of time and money according to the individuals' preferences and their bargaining power. The model is identified in the sense that both spouses' preferences and decision power, characterized by the Pareto weight, can be recovered from observed household behaviours.

Empirical implementation. We adopt a parametric version of the collective model for two reasons. First, the inclusion of multiple domestic production requires price variation for the home-produced goods. Despite the nonparametric approach is less restrictive, it cannot tackle this issue. Second, it yields a precise estimation of all parameters, which leads to fruitful discussions on all mechanisms within households. The utility functions and production functional forms we choose are flexible and fit in the literature on a parametric collective model by Cherchye et al. (2012).

We apply our model to time use and consumption data from the Longitudinal Internet Studies for the Social Sciences (LISS) in the Netherlands. An important feature of the data is that it includes information on the allocation of both time use and assignable goods within households. More importantly, LISS has direct measures on joint leisure time the spouses spend together, and the joint expenditure the spouses spend for different activities. This allows us to distinguish between four uses of time among married couples: market work, chores, childcare and leisure, the latter being divided into private and joint leisure; and also distinguish between four uses of expenditure: private consumption and family joint consumption on children, chores and joint experiences. The observation of household behaviours allows us to identify the model and investigate the trade-offs between time use and consumption in married couples with children.

Contribution & results. The main contributions of this paper are summarized into four. First, we apply a parametric collective model with household production in Cherchye et al. (2012) to account for the time and money investment in togetherness. The closest studies to ours, Browning et al. (2021) and Cosaert et al. (2023), estimate collective models with joint time. Browning et al. (2021) investigates the substitutability between spouses' private and joint leisure using Danish data and flexible functional specification, and finds that joint and private leisure are imperfect substitutes. Cosaert et al. (2023) considers various types of togetherness and allows household members to determine the timing of their market work. The paper presents that togetherness requires time synchronization using Dutch data and a nonparametric characterization. Compared to previous studies, we incorporate the money dimension of togetherness. We admit that the household expenditure spouses spend on their joint experience also contributes to their togetherness value and we explicitly model this. Therefore, we compare the marginal benefit of time and money, and in particular investigate whether time and money are substitutes or complements in producing the joy of togetherness. The result has important policy implications on spouses' working schedule synchronization, labor participation, marriage stability and children's development.

Second, we adopt a flexible parametric specification and structurally estimate the model despite its huge computational burden. Based on the estimation result, we present husbands' and wives' preferences over private leisure, private consumption and three home-produced goods: children's welfare, living environment and spouses' togetherness joy. We show how spouses' preferences and bargaining power influence the intrahousehold allocation of time and money. We find that, on average, wives attach higher marginal utilities to children's welfare, living environment and spouses' togetherness than men do. In the average household, husbands have slightly more bargaining power in household decision-making. The result indicates that wives' higher valuation on household production is the main source of their higher enrollment in unpaid household work, including childcare and chores.

Third, we allow the elasticity of substitution between joint leisure time and money to vary across households with different numbers of kids and years of relationship. We use these two variables because they significantly enter into the home production as production shifters, and thus enable the model identification. Moreover, the variation increases between-household heterogeneity. On average, the joint time and money households invest in togetherness are substitutes. The substitutability degree grows with the number of kids and shows an inverted-U shape correlation with the years of the relationship. The intuition is that spouses with more kids are more likely to substitute money and time for the joy of togetherness; while spouses having a short-term relationship are more likely to substitute time and consumption, but after reaching a certain year, the likelihood drops.

Fourth, we implement a series of counterfactual analyses for household behaviours. The main advantage of structural modelling is its effectiveness in counterfactual practices. We calculate the indifference scales of counterfactual separated couples to show their marriage gain from togetherness and economies of scale. We observe that the husband needs around 72% of the initial household resources to be as well off as when living alone, and the wife needs around 58%. The difference originates from wives' higher involvement in domestic productions and lower reliance on their husbands' time investment. Meanwhile, we show that spouses' togetherness value and children's welfare are positively related, this implies that parents' relationship and children's well-being are interconnected in families. Then, we show that married couples are more willing to increase their money investment rather than joint time for togetherness when their hourly wage increases.

Related literature. Since Gary Becker's treatise on marriage and family (Becker, 1973, 1974, 1991), household economics studies decisions made in the family. Empirical models of family economics have traditionally assumed a unitary household decision process in which the household behaves as a single decision-maker. The unitary model fails to describe intrahousehold allocations determined by the interaction of multiple individuals with different objects. An attractive alternative is the collective model proposed by Chiappori (1988, 1992) and Apps and Rees (1988). The collective model explicitly recognizes that the household consists of multiple individuals who each have their own rational preferences. These individuals are assumed to take Pareto efficient decisions that result from an intrahousehold bargaining process. Apps and Rees (1996) point out that the simple dichotomization of time into leisure and market work may distort welfare analysis. Chiappori (1997) then extend the framework to include home production, which incorporates time allocated to household activities such as chores and childcare. Blundell et al. (2005) further characterize home production as a domestic good produced by both household expenditure and individual time. Cherchye et al. (2012) generalize the collective model with home production to a setting with more than one domestic good: children's welfare and living environment, and provide new identification and estimation methods for empirical traceability. Our paper is a natural extension of the previous setting, by incorporating a new home production togetherness that is produced by couples' joint time and household expenditure.

Our focus on togetherness should not be surprising. There has long been evidence that couples like spending time together (Sullivan, 1996; Qi et al., 2017). Couples synchronize working schedules for more joint leisure within the family, which is evidence for togetherness preferences (Klaveren and Brink, 2007; Brown et al., 2011; Mansour and McKinnish, 2014). Sociological studies show that employees in countries with shorter working hours have more work-family conflicts (Cousins and Tang, 2004; Ruppanner and Maume, 2016). A series of studies suggest companionship has a positive correlation to marital happiness Palisi (1984); Agate et al. (2009); Ariyo and Mgbeokwii (2019). However, a sole focus on joint leisure time results in an overgeneralization of spouses' marriage satisfaction (Shebilske, 1999). Spouses have different subjective evaluations of togetherness (Crawford et al., 2002); and couples' experience of togetherness, e.g. quality of leisure involvement, is more important to their marital satisfaction (Johnson et al., 2006). Our paper will apply a collective model to estimate spouses' different preferences for togetherness value. We will model togetherness as the joint experience of spouses' interaction activities, taking into account both their joint leisure time and monetary expenditure.

Our interest in time and money investment in domestic production is essential for policy implications. Some studies examine the relation between time and monetary expenditure inputs into household-produced commodities by estimating their elasticity of substitution (Hamermesh, 2008; Gardes, 2019; Nevo and Wong, 2019). The substitution magnitude measures the money/time trade-offs and varies across commodities (Canelas et al., 2019). Policy proposals that ignore the time and money relation in home production may fail to achieve their objective, e.g. nutritional policies discussed by Davis (2014). Our paper will be the first to consider time and monetary investment in togetherness production within the household, from which we provide implications for labor policy and spouses' working synchronization. The result, whether joint leisure and money inputs are substitutes or complements, can enrich studies of time stress and time crunch among working couples, especially the couples with high earnings (Hamermesh and Lee, 2007).

The paper is organized as follows. Section 2 presents a collective model with home productions and shows the sketch of the identification proof. Section 3 provides the parametric specification of our model and discusses the estimation results. Section 4 implements several counterfactual analyses related to the research questions and section 5 concludes.

2 Theoretical Framework

2.1 Model setup

In this paper, we develop a collective labor supply model with household production in the spirit of Chiappori (1988, 1992); Blundell et al. (2005); Cherchye et al. (2012).

We focus on households with two adult members ($i = 1, 2$) whose preferences are represented by separate individual utility functions $\mathbf{U}^i$. Each adult spends time on private leisure l^i, market work m^i, childcare h_k^i, chores h_p^i and joint activities with their spouse l_t. The price of an individual time unit is set to the individual's working wages w^i, and the price of a joint time unit equals the sum of hourly wages by two members since the joint time has a sunk cost of both two adult's earnings. The household income, which consists of labor income and nonlabor income y, is allocated to individual private consumption c^i and family public consumption on childcare c^k, chores c^p and spouse's joint activities c^t. Each household has three forms of domestic production: the production of children's welfare u^k, the production of nice living conditions u^p and the production of joy from being together with their spouse u^t. The production process combines marketable goods and time spent by adult members and leads to nonmarketable public goods shared by adult members in the household. We also include a vector of production shifters s that move the production frontier while having no effect on preferences and income in the production function. The production technologies and shifters are different among different goods. Each individual gains utilities from private leisure, private consumption and the outputs of domestic production. The household maximizes the weighted sum of utilities from the two adult members, and the Pareto weight λ^i before each individual utility represents the bargaining position of the corresponding member. There are several factors that influence the Pareto weight: the relative wage of adult members w^1 and w^2, household nonlabor income y and other distribution factors z. We define the distribution factors as variables that have no impact on preferences and income so that endogeneity is not a concern in the model. The outcome of the utility maximization program is Pareto efficient and represents the optimal allocation of the intra-household bargaining process.

$$\max_{C} \lambda^1(w^1, w^2, y, z)\mathbf{U}^1(c^1, l^1, u^k, u^p, u^t) + \lambda^2(w^1, w^2, y, z)\mathbf{U}^2(c^2, l^2, u^k, u^p, u^t) \tag{1}$$

subject to

$$\begin{gathered}
u^k = u^k(c^k, h_k^1, h_k^2; s), \\
u^p = u^p(c^p, h_p^1, h_p^2; s), \\
u^t = u^t(c^t, l^t; s), \\
c^1 + c^2 + c^k + c^p + c^t = y + w^1 m^1 + w^2 m^2, \\
m^i + l^i + h_k^i + h_p^i + l^t = T \quad (i = 1, 2).
\end{gathered}$$

where $C = (c^i, l^i, c^k, h_k^i, c^p, h_p^i, c^t, l^t), \quad (i = 1, 2)$.

Following the previous literature on collective model (Chiappori, 1988, 1992; Blundell et al., 2005), we use two-stage allocation representation to solve the utility maximization program Eq.1. In the first stage, adult members both agree on certain levels of domestic outputs $\overline{u}^p$, $\overline{u}^k$, $\overline{u}^t$ and the conditional sharing rules ρ^i for each individual $i = 1, 2$, [1]

$$\rho^i(w^1, w^2, y, z, s) = w^i l^i(w^1, w^2, y, z, s) + c^i(w^1, w^2, y, z, s) - w^i.$$

[1] Conditional sharing rule is defined as the residual non-labour income after expenditures on the private commodity.

In this step, the household fixes the expenditure and time spent on each of the production.

In the second stage, each individual maximizes their own utility conditional on budget limits from the first stage,

$$\max_{c^i\, l^i} \mathbf{U}^i(c^i, l^i; \overline{u}^k, \overline{u}^p, \overline{u}^t) \tag{2}$$

subject to

$$c^i + w^i l^i = w^i + \rho^i.$$

and solve for the optimal allocation of private consumption and private leisure as functions of conditions set in the first stage. From this step, we get indirect utility functions for each individual.

We then substitute the result from the second step into the household maximization program Eq.1 to obtain an optimal allocation of other variables as follows

$$\max_{\rho^1,\rho^2,u^k,u^p,u^t} \lambda^1(w^1, w^2, y, z)\hat{v}^1(w^1, \rho^1, u^k, u^p, u^t) + \lambda^2(w^1, w^2, y, z)\hat{v}^2(w^2, \rho^2, u^k, u^p, u^t) \tag{3}$$

subject to

$$\rho^1 + \rho^2 + g^k(w^1, w^2)u^k + g^p(w^1, w^2)u^p + g^t(w^1, w^2)u^t = y.$$

Using the two-stage allocation representation, we obtain the optimal allocation of choice variables as smooth functions of the state variables including wages, nonlabor income, distribution factors and production shifters. Since the state variables are observed from the dataset, we can recover the household behaviors according to the smooth functions.

2.2 Identification

Now that we have obtained the observations of the household behaviors, the next step is to check whether the observation uniquely identifies the individual preferences and the Pareto weight. Our proof follows the framework shown in Cherchye et al. (2012), and extends the previous production functions in the inclusion of two public inputs in the same domestic production process. In the setup, we presume the price of each individual time unit to be the sunk cost of the equivalent working hour of the associated individual, and the price for the commodity to be one. In the production of children's welfare and good living conditions, the time efforts are individuals' separate investments, so that the price variation is distinguished by the wage rates. When the household spends time on joint activities with the spouse, the time inputs are joint and we show in the proof that the price of joint time equals the sum of wage rates by two spouses. The rest of the identification is mostly in line with the previous proof but only requires one additional significant production shifter, which is shown by the following proposition.

Proposition 1 *Let* $(c^i, l^i, c^k, c^p, c^t, h^i_k, h^i_p, l^t; i = 1, 2)$ *be observed functions of* $(w^1, w^2, y, \mathbf{z}, \mathbf{s})$ *and assume that there exists a distribution factor* z *and two production shifters* s_1, s_2 *such that* $\begin{bmatrix} \frac{\partial u^k}{\partial z} & \frac{\partial u^k}{\partial s_1} & \frac{\partial u^k}{\partial s_2} \\ \frac{\partial u^p}{\partial z} & \frac{\partial u^p}{\partial s_1} & \frac{\partial u^p}{\partial s_2} \\ \frac{\partial u^t}{\partial z} & \frac{\partial u^t}{\partial s_1} & \frac{\partial u^t}{\partial s_2} \end{bmatrix}$ *is invertible in a subset of the domain of* $u^j(w^1, w^2, y, z, s)$ *where* $j = k, p, t$, *then the individual preferences are generically identified.*

The detailed proof step is very technical so we put it in Appendix A and we only provide the general idea in the main text. The proof is a backward induction of the two-stage representation.

We start with the production functions. Since the production inputs are smooth functions of observed state variables, the input variables are also observed from the data. We then easily recover production functions u^j $(j = k, t, p)$ from input variables up to a strictly increasing transformation. To address the fundamental identification issue of differentiating individual preferences from production technologies, we introduce the usage of at least two production shifters and one distribution factor. Due to the variation provided by the significant shifters and factors, we are able to keep the domestic outputs constant while allowing variability in w^i and y, so that we disentangle the preferences and technology impacts of the same income and wages. Now that we have identified the production functions, we proceed with the second stage representation that each individual maximizes their own utility given

	Husband		Wife		Household		Obs
	Mean	SD	Mean	SD	Mean	SD	
Expenditures(euros per mouth)							
Private consumption	326.50	353.23	363.24	306.48			515
Childcare					418.32	370.41	515
Chores					1575.47	817.60	515
Joint activities					338.99	315.81	515
Time use(hrs per week)							
Market labor	47.66	12.01	27.72	12.72			515
Private leisure	91.43	17.13	95.45	17.81			515
Childcare	8.80	8.68	14.24	14.38			515
Chores	11.10	8.06	21.38	11.99			515
Joint activities					9.91	7.57	515
Demographics							
Age	43.95	6.59	41.85	6.68			515
Wage rate	12.66	4.85	10.96	4.35			515
Household nonlabor income					99.81	455.39	515
Number of kids					2.04	0.74	515
Mean age of kids					11.79	5.95	515
Year of relationship					18.50	7.37	515

Table 1: Summary Statistics

the fixed output level and conditional sharing rules. The second stage formulation shows that we can recover the individual preferences and the sharing rule up to an additive constant for given presumptions(Blundell et al., 2005). The last and crucial step of the proof is to show the uniqueness of the transformation, which indicates the one-to-one mapping between household behavior observations and individual preferences and the Pareto weight. We implement the uniqueness proof by counter-assuming two tuples generate the same solutions to the utility maximization program. We show that the transformations associated with the two tuples are identical so that all parts of the model are generically identified.

3 Empirical Application

3.1 Data overview

We apply our model to a sample of households drawn from the Longitudinal Internet Studies for the Social Sciences Panel (hereafter LISS) covering the years 2009, 2010 and 2012. We combine questionnaires from different modules: (1) Household box for background variables; (2) Core study 5 Family and Household; (3) Core study 6 Work and Schooling; (4) Assembled study 34 Time Use and Consumption. The assembled study provides for each household detailed information about consumption and time-use.

The selection of the sample households conforms to several rules due to the nature of the research questions. We focus on couples both at their prime ages between 25 to 60 and participating in the labor market. The couples are in the heterosexual marriage status and have at least one living-at-home child below 18 years old. After data cleaning, we obtain a sample of 515 households for our further estimation.

We have demographics including adult members' ages, wage rates, household non-labour income, their years of relationship, number of kids, the age of each kid, and detailed time use and consumption including the key variables 'joint leisure time' defined as the leisure time spent together with a partner, and 'consumption on joint activities with the spouse' defined as expenditure on eating at home, trips and holidays with a partner. The detailed variable formulation is included in Appendix B.

The summary statistics are as follows. The households in our sample spend on average 2332.78 euros each month on household public goods, explicitly 418.32 euros on childcare, 1575.47 euros on household chores and 338.99 euros on joint activities with partners. The females spend on average 363.24 euros per month on their personal matters which is higher than the male's at 326.50 per month. Although the expenditure on togetherness is not comparable to the expense on children and housework, it's still higher than some private spending and takes up 14.53% of the average family's public expense. Time usage is mostly divided into individual calculations. Females in our sample spend 27.72 hours weekly on market work(commuting time included), while the average working time for males is around 42% higher at 47.66 hours per week. The females and males spend on average 14.24 hours and 8.80 hours on childcare, separately. And the time for chores is 21.38 and 11.10 corresponding for females and males. It is not surprising that females take more responsibilities in childcare and chores. The couples spend on average 9.91 hours on joint leisure activities each week. Although the average joint leisure takes up a small percentage compared to other time divisions, it is still higher than male childcare time.

Apart from the division of time and expenses, females and males also present distinguishing differences in the demographics. Males' average hourly wage is 1.7 euros higher than that of females, and the males in the sample have an average of 1.9 years of higher age. From the perspective of households as an integral, each family has on average 2 kids at home and the children's mean age is 11.79 years old. In the sample selection, we only count the children as living at home if their age is below 18 because we deem 18 to be the independent age and the parents spare fewer duties after a child turns 18. In terms of the relationship, the spouses have an average of 18.5 years of relationship in our sample.

3.2 Parametric specification

To implement the empirical application, we introduce the parametric specification for the representation of the adult members' preferences and domestic production.

Due to the nature of the two-stage allocation representation, we only need to assume individuals' indirect utility functions for the straightforward derivation of private consumption and private leisure using Roy's identity. Hence, individuals' preferences concerning their own consumption and leisure conditional on the fixed domestic outputs can be shown as:

$$v^i\left(w^i, \rho^i, \bar{u}^k, \bar{u}^p, \bar{u}^t\right) = \frac{\ln\left(w^i + \rho^i\right) - \ln a^i\left(w^i, \bar{u}^k, \bar{u}^p, \bar{u}^t\right)}{\left(w^i\right)^{\beta^i}}, \tag{4}$$

where

$$\ln a^i\left(w^i, \bar{u}^k, \bar{u}^p, \bar{u}^t\right) = \left(a_1^i(\mathbf{d}^i) + a_2^i \ln \bar{u}^k + a_3^i \ln \bar{u}^p + a_4^i \ln \bar{u}^t\right) \ln w^i.$$

The $\mathbf{d}^i$ is a vector of individual taste shifters, in our paper only age is included so we have

$$a_1^i(\mathbf{d}^i) = a_{10}^i + a_{11}^i * age^i (i = 1, 2)$$

The indirect utility functions are of the Price-Independent Generalized Logarithmic (hereafter PIGLOG) form with normalized price for Hicksian composite commodity equals 1, so the specification coincides with the Almost Ideal Demand System and can be used to test the restrictions of the utility maximization program. We take the PIGLOG class because it ensures the aggregate demands be written in the same form where the individuals' total budget is replaced with the sum of expenditure on private leisure and the Hicksian composite commodity.

The produced domestic products are for public sharing between the spouses, but each spouse attaches a different value to the same goods. We assume the production technologies have the forms of constant elasticity of substitution and with constant returns to scale as below. For $j = (k, p)$,

$$u^j\left(c^j, h_j^1, h_j^2; \mathbf{s}^j\right) = \left(\gamma_j^1\left(h_j^1\right)^{\epsilon^j\left(\mathbf{s}^j\right)} + \gamma_j^2\left(h_j^2\right)^{\epsilon^j\left(\mathbf{s}^j\right)} + \gamma_j^3\left(c^j\right)^{\epsilon^j\left(\mathbf{s}^j\right)}\right)^{\frac{1}{\epsilon^j\left(\mathbf{s}^j\right)}},$$

and for $j = t$,

$$u^j\left(c^j, l^j; \mathbf{s}^j\right) = \left(\gamma_j^1\left(l^j\right)^{\epsilon^j\left(\mathbf{s}^j\right)} + \gamma_j^2\left(c^j\right)^{\epsilon^j\left(\mathbf{s}^j\right)}\right)^{\frac{1}{\epsilon^j\left(\mathbf{s}^j\right)}}.$$

where $\epsilon^j(\mathbf{s}^j)$ is assumed to depend on the production shifters in $\mathbf{s}^j$.

We also assume the price indexes for the produced goods are of corresponding CES form. For $j = (k, p)$,

$$g^j\left(w^1,w^2\right)=\left[\left(\gamma_j^1\right)^{\frac{-1}{\epsilon^j\left(\mathbf{s}^j\right)-1}}\left(w^1\right)^{\frac{\epsilon^j\left(\mathbf{s}^j\right)}{\epsilon^j\left(\mathbf{s}^j\right)-1}}+\left(\gamma_j^2\right)^{\frac{-1}{\epsilon^j\left(\mathbf{s}^j\right)-1}}\left(w^2\right)^{\frac{\epsilon^j\left(\mathbf{s}^j\right)}{\epsilon^j\left(\mathbf{s}^j\right)-1}}+\left(\gamma_j^3\right)^{\frac{-1}{\epsilon^j\left(\mathbf{s}^j\right)-1}}\right]^{\frac{\epsilon^j\left(\mathbf{s}^j\right)-1}{\epsilon^j\left(\mathbf{s}^j\right)}},$$

and for $j=t$,

$$g^j\left(w^1,w^2\right)=\left[\left(\gamma_j^1\right)^{\frac{-1}{\epsilon^j\left(\mathbf{s}^j\right)-1}}\left(w^1+w^2\right)^{\frac{\epsilon^j\left(\mathbf{s}^j\right)}{\epsilon^j\left(\mathbf{s}^j\right)-1}}+\left(\gamma_j^2\right)^{\frac{-1}{\epsilon^j\left(\mathbf{s}^j\right)-1}}\right]^{\frac{\epsilon^j\left(\mathbf{s}^j\right)-1}{\epsilon^j\left(\mathbf{s}^j\right)}}.$$

To allow for some variation in production technologies across households, we incorporate a vector of production shifters $\mathbf{s}$. The shifters are independent of personal preferences and household income but shift the production-possibility frontiers so that we avoid endogeneity issues. The heterogeneous effects shown by production shifters allow us to differentiate the production variation and individual preferences for the change of wage and income. However, shifters enter into production functions discrepantly as below. For $j=(k,p)$,

$$\epsilon^j(s^j)=\epsilon_0^j+\epsilon_1^j*\text{number of kids}+\epsilon_2^j*\text{years of relationship},$$

and for $j=t$,

$$\epsilon^j(s^j)=\epsilon_0^j+\epsilon_1^j*\text{number of kids}+\epsilon_2^j*\text{years of relationship}+\epsilon_3^j*\text{years of relationship}^2.$$

Since the production function for u^k and u^p are CES forms of three inputs, the elasticity of substitution between pairs of factors be identical (Uzawa, 1962). For u^t the elasticity of substitution only shows the substitutability or complementarity between time and expenditure inputs on joint activities.

We also assume that the Pareto weight is of exponential form ranging between 0 and 1.

The Pareto weight depends on the spouses' relative wage, household nonlabor income and a vector of distribution factors, the latter of which do not impact individual preferences and the budget constraint for the endogeneity concern. Cherchye et al. (2012) have shown that age differences influence the partners' bargaining power within households in the Netherlands so we follow and adopt this variable in our application of Dutch households. We can of course include more distribution factors, but it's not the main idea of this study and increases the estimation difficulty. According to Proposition 2.1, one distribution factor and two production shifters are sufficient for model identification.

$$\lambda^1(w^1, w^2, y, \mathbf{z}) = \frac{exp(\Lambda_1 + \Lambda_2 \frac{w^1}{w^2} + \Lambda_3 y + \Lambda_4' \mathbf{z})}{1 + exp(\Lambda_1 + \Lambda_2 \frac{w^1}{w^2} + \Lambda_3 y + \Lambda_4' \mathbf{z})}.$$

Following the two-stage allocation rules we show in the model setup, we solve the model using the parametric specifications in steps.

We start from the second stage where each adult member chooses optimal private consumption and private leisure conditional on given domestic goods and conditional sharing rules. The utility maximization enables us to apply Roy's identity to the indirect utility functions and get the Marshallian demand function of private leisure for $i = (1, 2)$ as:

$$l^i = -\frac{\partial v^i / \partial w^i}{\partial v^i / \left(w^i + \rho^i\right)} = \frac{\left(w^i + p^i\right)}{w^i} \left[a_1^i \left(d^i\right) + a_2^i \ln \bar{u}^k + a_3^i \ln \bar{u}^p + a_4^i \ln \bar{u}^t + \beta^i \ln \frac{w^i + p^i}{a^i \left(w_i; \bar{u}^k, \bar{u}^p, \bar{u}^t\right)}\right] \tag{5}$$

based on the personal budget constraint $c^i = w^i + \rho^i - w^i l^i$, we also get the Marshallian demand function of private consumption as:

$$c^i = \left(w^i + \rho^i\right) \left[\left(1 - a_1^i \left(d^i\right) - a_2^i \ln \bar{u}^k - a_3^i \ln \bar{u}^p - a_4^i \ln \bar{u}^t\right) - \beta^i \ln \left(\frac{w^i + \rho^i}{a^i(w^i; \ln \bar{u}^k, \ln \bar{u}^p, \ln \bar{u}^t}\right)\right] \tag{6}$$

We then proceed to the first stage where the two adult members collectively decide the resources invested in home production and the resources for personal use. We formulate the household utility maximization program as

$$\max_{\rho^1,\rho^2,u^k,u^p,u^t} \lambda(w^1,w^2,y,\mathbf{z})\left(\frac{\ln\left(w^1+\rho^1\right)-\ln a^1\left(w^1;u^k,u^p,u^t\right)}{(w^1)^{\beta 1}}\right) \\ +(1-\lambda(w^1,w^2,y,\mathbf{z}))\left(\frac{\ln\left(w^2+\rho^2\right)-\ln a^2\left(w^2;u^k,u^p,u^t\right)}{(w^2)^{\beta 2}}\right) \tag{7}$$

subject to

$$\rho^1+\rho^2+g^k\left(w^1,w^2\right)u^k+g^p\left(w^1,w^2\right)u^p+g^t\left(w^1,w^2\right)u^t=y.$$

where the $g^j\left(w^1,w^2\right)$ are price index for produced goods $j=(k,p,t)$.

Assuming an interior solution, we derive the first-order condition for each choice variable and rewrite them as:

$$\begin{aligned}
w^1+\rho^1 &= \frac{1}{\mu}\frac{\lambda}{(w^1)^{\beta 1}} \\
w^2+\rho^2 &= \frac{1}{\mu}\frac{(1-\lambda)}{(w^2)^{\beta 2}} \\
g^k\left(w^1,w^2\right)u^k &= \frac{1}{\mu}\left[-\frac{\lambda}{(w^1)^{\beta 1}}\alpha_2^1\ln w^1-\frac{(1-\lambda)}{(w^2)^{\beta 2}}\alpha_2^2\ln w^2\right] \\
g^p\left(w^1,w^2\right)u^p &= \frac{1}{\mu}\left[-\frac{\lambda}{(w^1)^{\beta 1}}\alpha_3^1\ln w^1-\frac{(1-\lambda)}{(w^2)^{\beta 2}}\alpha_3^2\ln w^2\right] \\
g^t\left(w^1,w^2\right)u^t &= \frac{1}{\mu}\left[-\frac{\lambda}{(w^1)^{\beta 1}}\alpha_4^1\ln w^1-\frac{(1-\lambda)}{(w^2)^{\beta 2}}\alpha_4^2\ln w^2\right]
\end{aligned} \tag{8}$$

.

Adding up these equations we have the budget constraint on the left-hand side and derive an expression for the shadow price:

$$\mu=\frac{X(w^1,w^2,\lambda)}{w^1+w^2+y} \tag{9}$$

where $X(w^1,w^2,\lambda) = \frac{\lambda}{(w^1)^{\beta 1}}(1-(\alpha_2^1+\alpha_3^1+\alpha_4^1)\ln w^1) + \frac{(1-\lambda)}{(w^2)^{\beta 2}}(1-(\alpha_2^2+\alpha_3^2+\alpha_4^2)\ln w^2)$.

Plugging the shadow price back to Eq.8, we get production functions w^j and $j = (k, p, t)$ consistent with the collective rationality. Now we have two key calculations. First, we put the production functions derived from the household utility maximization program back to the individual conditional optima Eq.5 and Eq.6, then we obtain the optimal allocation for private consumption and private leisure:

$$\begin{aligned} l^i &= \frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\frac{\lambda^i}{(w^i)^{\beta^i}}\frac{1}{w^i}\left[A^i+\beta^i\left(\ln\frac{(w^1+w^2+y)\lambda^i}{X(w^1,w^2,\lambda)(w^i)^{\beta^i}}-A^i\ln w^i\right)\right] \\ c^i &= \frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\frac{\lambda^i}{(w^i)^{\beta^i}}\left[1-A^i-\beta^i\left(\ln\frac{(w^1+w^2+y)\lambda^i}{X(w^1,w^2,\lambda)(w^i)^{\beta^i}}-A^i\ln w^i\right)\right] \end{aligned} \tag{10}$$

Second, we plug the shadow price back into Eq.8 and then obtain the separate cost functions for the three produced goods. We assume the production processes are cost-minimizing so that we can apply Shephard's lemma to the cost functions and derive the optimal allocation for all the inputs:

$$\begin{aligned}
h_k^1 &= \left(\frac{w^1}{\gamma_k^1}\right)^{\frac{1}{\varepsilon^k(s^k)-1}}\left(g^k(w^1,w^2)\right)^{-\frac{\varepsilon^k(s^k)}{\varepsilon^k(s^k)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_2^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_2^2\ln w^2\right] \\
h_k^2 &= \left(\frac{w^2}{\gamma_k^2}\right)^{\frac{1}{\varepsilon^k(s^k)-1}}\left(g^k(w^1,w^2)\right)^{-\frac{\varepsilon^k(s^k)}{\varepsilon^k(s^k)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_2^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_2^2\ln w^2\right] \\
c^k &= \left(\frac{1}{\gamma_k^3}\right)^{\frac{1}{\varepsilon^k(s^k)-1}}\left(g^k(w^1,w^2)\right)^{-\frac{\varepsilon^k(s^k)}{\varepsilon^k(s^k)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_2^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_2^2\ln w^2\right] \\
h_p^1 &= \left(\frac{w^1}{\gamma_p^1}\right)^{\frac{1}{\varepsilon^p(s^p)-1}}\left(g^p(w^1,w^2)\right)^{-\frac{\varepsilon^p(s^p)}{\varepsilon^p(s^p)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_3^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_3^2\ln w^2\right] \\
h_p^2 &= \left(\frac{w^2}{\gamma_p^2}\right)^{\frac{1}{\varepsilon^p(s^p)-1}}\left(g^p(w^1,w^2)\right)^{-\frac{\varepsilon^p(s^p)}{\varepsilon^p(s^p)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_3^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_3^2\ln w^2\right] \\
c^p &= \left(\frac{1}{\gamma_p^3}\right)^{\frac{1}{\varepsilon^p(s^p)-1}}\left(g^p(w^1,w^2)\right)^{-\frac{\varepsilon^p(s^p)}{\varepsilon^p(s^p)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_3^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_3^2\ln w^2\right] \\
l^t &= \left(\frac{w^1+w^2}{\gamma_t^1}\right)^{\frac{1}{\varepsilon^t(s^t)-1}}\left(g^t(w^1,w^2)\right)^{-\frac{\varepsilon^t(s^t)}{\varepsilon^t(s^t)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_4^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_4^2\ln w^2\right] \\
c^t &= \left(\frac{1}{\gamma_t^2}\right)^{\frac{1}{\varepsilon^t(s^t)-1}}\left(g^t(w^1,w^2)\right)^{-\frac{\varepsilon^t(s^t)}{\varepsilon^t(s^t)-1}}\frac{(w^1+w^2+y)}{X(w^1,w^2,\lambda)}\left[-\frac{\lambda}{(w^1)^{\beta^1}}\alpha_4^1\ln w^1-\frac{1-\lambda}{(w^2)^{\beta^2}}\alpha_4^2\ln w^2\right]
\end{aligned} \tag{11}$$

where

$$X(w^1, w^2, \lambda) = \frac{\lambda^1}{(w^1)^{\beta^1}} \left(1 - \left(\alpha_2^1 + \alpha_3^1 + \alpha_4^1\right) \ln w^1\right) + \frac{(1-\lambda^1)}{(w^2)^{\beta^2}} \left(1 - \left(\alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) \ln w^2\right)$$

and

$$\begin{aligned} A^i \equiv & \alpha_1^i \left(\mathbf{d}^i\right) + \alpha_2^i \ln\left(\frac{w^1+w^2+y}{X\left(w^1,w^2,\lambda\right)} \frac{1}{g^k\left(w^1,w^2\right)} \left[-\frac{\lambda^1}{\left(w^1\right)^{\beta^1}} \alpha_2^1 \ln w^1 - \frac{\left(1-\lambda^1\right)}{\left(w^2\right)^{\beta^2}} \alpha_2^2 \ln w^2\right]\right) \\ & + \alpha_3^i \ln\left(\frac{w^1+w^2+y}{X\left(w^1,w^2,\lambda\right)} \frac{1}{g^p\left(w^1,w^2\right)} \left[-\frac{\lambda^1}{\left(w^1\right)^{\beta^1}} \alpha_3^1 \ln w^1 - \frac{\left(1-\lambda^1\right)}{\left(w^2\right)^{\beta^2}} \alpha_3^2 \ln w^2\right]\right) \\ & + \alpha_4^i \ln\left(\frac{w^1+w^2+y}{X\left(w^1,w^2,\lambda\right)} \frac{1}{g^t\left(w^1,w^2\right)} \left[-\frac{\lambda^1}{\left(w^1\right)^{\beta^1}} \alpha_4^1 \ln w^1 - \frac{\left(1-\lambda^1\right)}{\left(w^2\right)^{\beta^2}} \alpha_4^2 \ln w^2\right]\right) \end{aligned}$$

Now we have the closed-form solutions for all choice variables as smooth functions of wages, nonlabor income, ages, years of relationship, the mean age of kids and the number of kids. The dependent variables are all observed from the data so we can also calculate the choice variables from observable.

Moreover, we also observe real data of all time and consumption variables from the sample. To reduce the distance between model solutions and observations, we minimize the sum of squared residuals over parameters and use a non-linear least square estimator for the estimation.

3.3 Empirical Prediction

Despite the highly nonlinear closed-form expressions shown in Eq.10 and Eq.11, we can still observe the mechanisms underlying the household behaviours and we use this framework to interpret the channels that impact the time and money allocation within households.

We examine the impacts of demographics on household behaviours. Age, as a taste shifter that varies with an individual's preference, decreases private consumption and private leisure but doesn't directly affect other time and consumption behaviours. The production shifters, number of kids, mean age of kids and the year of relationship, impact time and consumption invested in domestic production through shifting the production frontier and have no direct effects on private consumption and private leisure. The separate effects of taste shifters and production shifters are consistent with the spirit of modelling. We assume that the household decision-making process is divided into two stages. The household first agrees on a certain degree of domestic production outputs and then the spouses maximize their individual utilities conditional on the given outputs of domestic production. To disentangle the effects of certain variables through production mechanisms and preferences, we let production shifters be variables influencing household behaviours only through production technologies but not the individual preferences, and the taster shifters are variables impacting only individual preferences. As a result, the production shifters only directly influence the time and consumption allocation related to production, including childcare, chores and togetherness; while the taster shifter only has a direct effect on private consumption and private leisure. However, there are still indirect effects, to be more specific, the time and money allocations on private and production activities are still correlated due to the same budget and time constraints fixed in the household. For instance, the age taste shifter decreases one's expenditure on private consumption and private leisure as a direct effect; this simultaneously increases the constraints of time and money spent on production. Although the increasing effect varies across childcare, chores and togetherness activities, we can ambiguously find an indirect effect of taste shifter on production allocation. The same applies to the production shifters' indirect effects on private leisure and private consumption. Due to similar identification purposes, we let distribution factors be variables that have no effects on budget constraints and individual preferences, like relative wages, age differences and non-labour income. The distribution factors impact all choice variables through bargaining weight, but the exact allocation outcome depends on the interaction of bargaining weight, preferences over corresponding behaviours and production. Take a female's childcare time h_k^2 as an example, when the female's relative wage increases, she has more bargaining power

$(1-\lambda)$ while her spouse has lower bargaining power λ, but the impact outcome depends on the relative change of $\lambda\alpha_2^1(\ln w^1)/(w^1)^{\beta^1}$ and $(1-\lambda)\alpha_2^2(\ln w^2)/(w^2)^{\beta^2}$. If the female has a high attachment to childcare α_2^2, an increased but still low wage w_2, she will increase her childcare time despite higher bargaining power; if the female has a low valuation on childcare, a high wage, then she will reduce her childcare time with a higher bargaining power between couples.

We also examine the factor that leads to unequal time division on unpaid domestic work, including childcare and chores. Comparing the spouses' childcare time equation h_k^1 with h_k^2, we find the time difference comes from the couple's wage and childcare productivity per unit time w^i/γ_k^i [2]. This indicates that a spouse who has higher childcare productivity and a lower wage spends more time on childcare. The spouse who has higher childcare productivity is more efficient in childcare activities in the same amount of time, while the spouse who has a lower wage has lower sunk costs of reducing market work and specializing in unpaid household tasks. The reasoning lies consistent with household specialization and comparative advantage. The same goes for the analysis of household chores by comparing h_p^1 with h_p^2.

We also examine the togetherness within households. The contribution of time and money investment on togetherness value is represented as γ_t^1 and γ_t^2 separately, while the substitutability or complementarity between the two inputs is represented as $1/(1-\epsilon^t)$. The time and money input levels are determined by the couples' valuation of togetherness α_4^i, the output of the togetherness joy, and bargaining. Couples both enjoy togetherness but to different degrees, so the bargaining power will lead the investment to be more aligned with the preference of the spouse who has more bargaining power.

Due to the complex forms, some mechanisms are hard to interpret when analyzing the effects of a single factor, such as the impacts of individual wages. However, we will make full use of the structural model and implement a series of counterfactual analyses in Section

[2]since the elasticity ϵ^k falls in $(0,1)$, the wage/productivity ratio is negatively correlated to the time spent on children

Preferences	Estimate	SD	Production	Estimate	SD	Pareto weight	Estimate	SD
a_{10}^1	0.850	0.035	γ_k^1	0.280	0.011	λ_1	-0.323	0.067
$a_{11}^1[age^1/10]$	-0.012	0.007	γ_k^2	0.540	0.016	$\lambda_2[w_1/w_2]$	0.879	0.022
$a_2^1[u^k]$	-3.500	0.390	γ_k^3	0.100	0.006	$\lambda_3[y]$	-0.654	0.135
$a_3^1[u^p]$	-3.500	0.244	ϵ_0^k	-0.020	0.005	$\lambda_4[(age^1 - age^2)/age^2]$	-0.018	0.056
$a_4^1[u^t]$	-4.000	0.862	$\epsilon_1^k[kids]$	0.005	0.001			
β^1	0.210	0.010	$\epsilon_2^k[meanagekids/10]$	-0.002	0.000			
a_{10}^2	0.654	0.114	γ_p^1	0.278	0.001			
$a_{11}^2[age^2/10]$	0.014	0.005	γ_p^2	0.269	0.001			
$a_2^2[u^k]$	-2.140	0.100	γ_p^3	0.452	0.001			
$a_3^2[u^p]$	-3.000	0.353	ϵ_0^p	-0.010	0.000			
$a_4^2[u^t]$	-3.121	0.493	$\epsilon_1^p[kids]$	0.002	0.000			
β^2	0.066	0.021	$\epsilon_2^p[meanagekids/10]$	0.001	0.000			
			γ_t^1	0.976	0.002			
			γ_t^2	0.024	0.002			
			ϵ_0^t	0.239	0.003			
			$\epsilon_1^t[kids]$	0.200	0.002			
			$\epsilon_2^t[years/10]$	0.192	0.000			
			$\epsilon_3^t[yearsq/100]$	-0.152	0.000			

Table 2: Estimation result

4. The result of counterfactual analyses will give us more insights into the dynamics of bargaining channels, the tradeoff between time and money investment in togetherness, the marriage gain from togetherness etc.

3.4 Estimation results

First about the individuals' preference parameters. All three domestic goods have significant positive effects on both couples' individual utility. Women attach a higher marginal utility to children's welfare, chores, and togetherness joy than men do (see the estimates $\tilde{a}_2^1$, $\tilde{a}_3^1$, $\tilde{a}_4^1$, and $\tilde{a}_2^2$, $\tilde{a}_3^2$, $\tilde{a}_4^2$). This indicates women value domestic production, including togetherness values more than men within households given the same level of production values.

Second, about the production parameters. As for production technologies, we find that one extra time spent on children by the mother benefits more than one extra time spent on children by the father ($\gamma_k^2 > \gamma_k^1$). An opposite pattern emerges for the household chores ($\gamma_p^1 > \gamma_p^2$). For togetherness, one extra hour spent with spouses benefits togetherness more than 250 euros spent on joint activities with the spouse per week ($\gamma_t^1 > \gamma_t^2$). At given production technologies, the interaction between different inputs can be seen from the calculation of the elasticity of substitution for each domestic production good, which equals to $\frac{1}{1-\epsilon^j(s^j)}$ according to the parametric specification of production functions. For an average household, the elasticity of substitution between inputs for children's welfare is 0.96, which indicates that the time spent by the father, the time spent by the mother, and the expenditure on children are mutually complemented. For chores, the average elasticity of substitution amounts to 1.02, which shows that inputs for chores present weak substitutability. The average elasticity of substitution between togetherness expenditure and togetherness time is 1.38 with a fairly high standard deviation of 0.72, from which we conclude that the substitutability for togetherness inputs has a large heterogeneity across households. We further find that the substitutability magnitude for togetherness grows with the number of kids and presents an inverted-U shape correlation with the year of the relationship (see estimates of ϵ_1^t, ϵ_2^t and ϵ_3^t). The economic intuition is that households with more kids are more likely to substitute expenditure and time if at least one of their wages increases. Married couples who have a short-term relationship are more likely to substitute time and consumption in producing togetherness joy, however after reaching a certain year, they become less likely to swap time and consumption on the joint activities.

We can also interpret the estimation at the dimension of Pareto weight. The increasing nonlabor household income and increasing spousal age difference leads to decreasing bargaining power of males ($\lambda_3 < 0$, $\lambda_4 < 0$); while the higher relative wage leads to higher bargaining power of males ($\lambda_2 > 0$).

However, since the equation systems are highly non-linear and have 34 parameters, some effects are hard to interpret because too many factors should be considered via parameters and equations. Instead, we use counterfactual analysis and graphs to carry on the analysis in the following subsection.

4 Counterfactual Analysis

The following analyses reveal the complex interactions of three different mechanisms, individual preferences, bargaining power and the household's production technologies.

4.1 The effects of changing wages on intrahousehold allocation

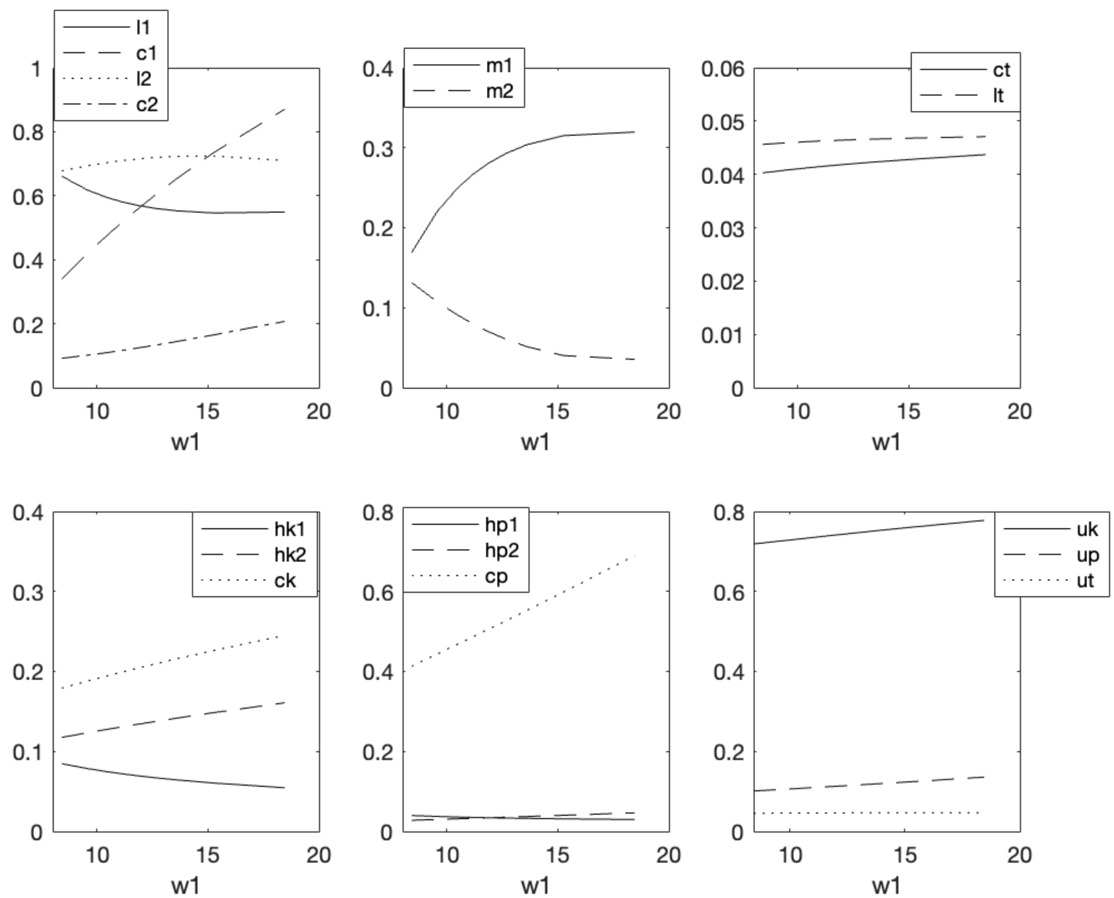

Figure 1: Effects of changing male wages

This subsection illustrates the effects of wage changes on the intrahousehold allocation of consumption and time use and the effects on domestic production.

Figure 1 shows the impacts of change in the husband's wage. The male wage change runs from the 1st decile to the 10th decile in the male wage distribution, while other explanatory variables are fixed at their means including the female wage.

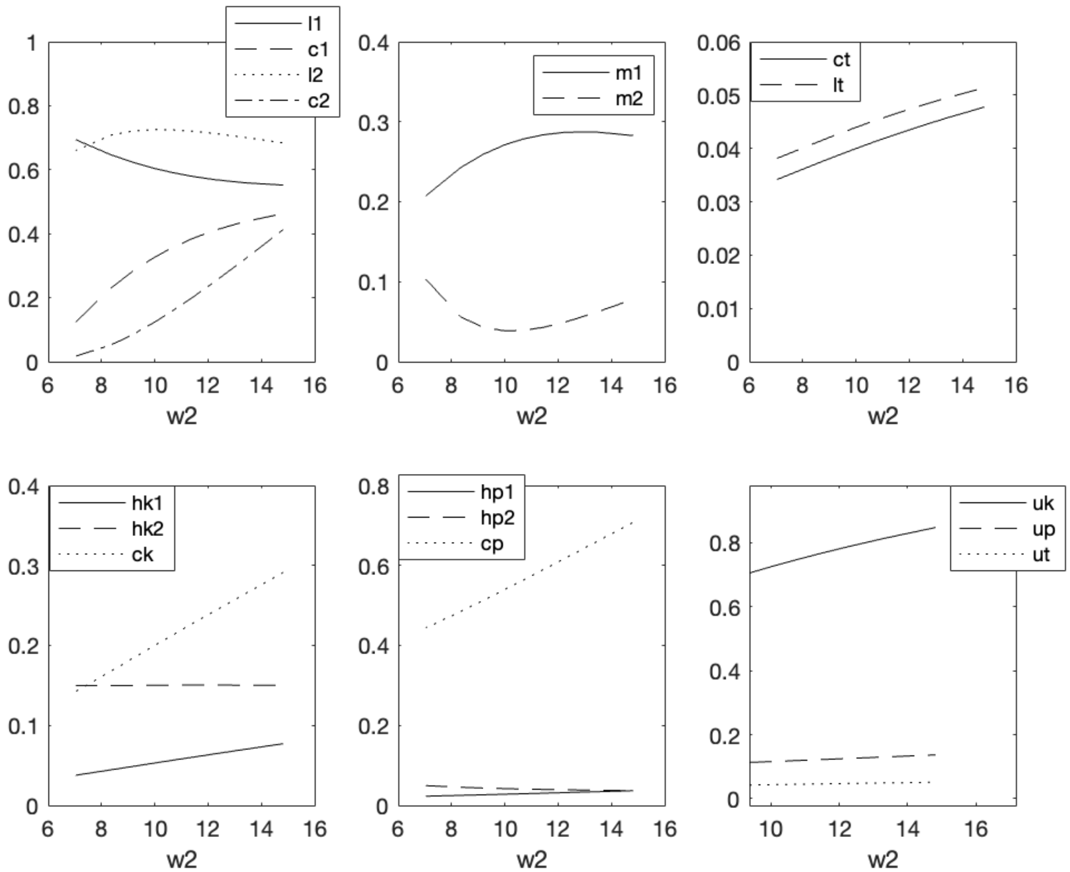

Figure 2: Effects of changing female wages

When the husband's wage increases, his market work increases, while this increase is accompanied by a decrease in his private leisure and in his time spent on childcare and household chores. The picture emerges differently when we look at the impacts on the female-dependent variables. When the husband's wage increases, the female market hours decrease while her time spent on housework and children increase. These increases compensate for the male's decrease in time spent on domestic goods.

Since the magnitude of the husband's working hours increase is larger than the wife's market work decline, the total household income increases as a consequence. This explains the rising trend of all consumption categories, even females' private consumption.

The consumption and time spent on togetherness both increase, but the increase in consumption is steeper than in joint leisure. This is in line with the previous statement that time and consumption are gross substitute inputs in producing the togetherness value for an average household. When the male wage increases, the household is more willing to increase its money than joint time invested in the production of togetherness joy.

Figure 2 shows the impacts of increasing female wages on dependent variables while keeping other explanatory variables fixed at their average level including the husband's wage. The pattern is somewhat different from Figure 1. When the female wage increases, the female labor supply first decrease, and only increases again when her wage reaches beyond the mean wage at 10. The female's private leisure responds oppositely to the market hour in that it initially increases and then slightly decreases. In the meanwhile, the male market work increases when the female wage increases but at a decreasing rate when above the average wage.

The pattern for domestic production is also different from Figure 1, especially in the time spent on childcare. When the female wage increases, both two spouses' childcare time increases while the wife's increases at a slower rate. But for chores, the male's time investment rises while the female's time spent declines. As for consumption, the rise in the female wage also leads to an increase in all consumption categories.

The joint leisure time and consumption of togetherness both increase in response to the higher female wage. Although it's not obvious from the figure, the increasing rate of consumption is higher than that of time, which once again shows higher substitutability between time and consumption inputs.

4.2 The effects of bargaining powers on domestic productions

An interesting discussion about households is which bargaining structure is more beneficial to domestic production. We see from the estimation result that the relative wage is positively correlated with a higher bargaining power of males and negatively related to the decision power of females at home. And the person with a higher bargaining position has

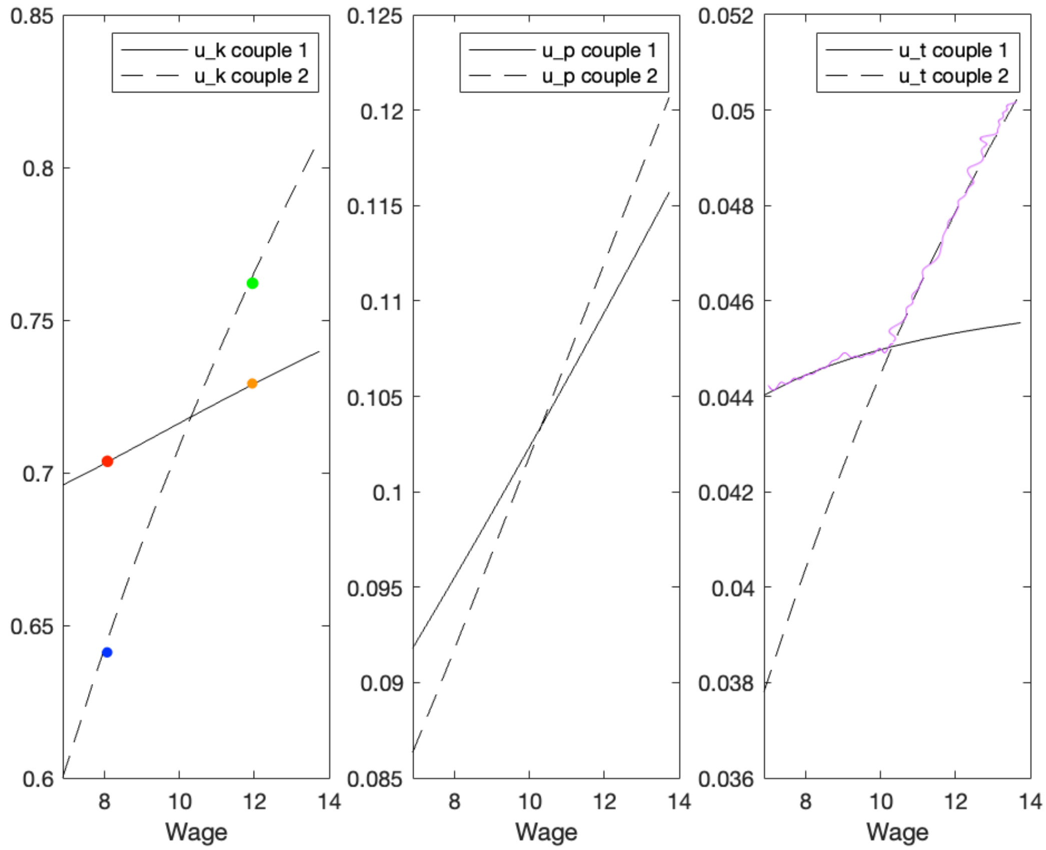

Figure 3: Effects of different bargaining positions

more discretion in intrahousehold allocation and their preference will be consistent with the household allocation.

In this part, we show whose decision power is more beneficial to domestic production. In the comparison, we focus on the average household by fixing the couples' characteristics to the mean values except for the wage rates. We have two types of couples in the analysis. In the first type, the male wage runs from the 1st decile to the 10th decile in the male wage distribution, while keeping the female wage to the average wage of all samples (10 euros per hour). When the changing male wage is below 10, the relative wage of the male is below 1 and the mother has more bargaining power within households; when the changing male wage is above 10, the father has more decisive power. In the second type, we change the female wage according to the distribution and keep the male wage to average at 10 euros per hour, and thus the father has higher bargaining power when the changing wage is below 10 and the mother has more discretion when the changing wage is above 10.

Take u^k as an example, the red dot shows the male has a wage rate of 7 and his spouse has an average wage rate of 10 (mother in power) and the blue dot shows the female has a wage rate of 10 and her spouse has an average rate of 10 (father in power). The red dot has a higher value of u^k, which indicates that the children's welfare is higher when the mother has higher bargaining power at home. The same case holds when the changing wage is above average, where the green dot shows that the mother is superior in bargaining position and the orange dot shows that the father has more decisive power. We compare the output level produced by the dots at the same vertical line and reach the conclusion that the average household achieves a higher children's welfare when the mother has more bargaining power at home. The argument holds the same for the other two domestic goods, and the effect is most obvious in the togetherness value [3].

4.3 Indifference scale

The indifference scale measures how much income an individual living alone needs to have in order to be as well off as when living in a couple with some given household income. Cherchye et al. (2012) calculated the indifference scale of households with two domestic goods, children's welfare and household chores. In this paper, we observe and compare the minimum income needed to be as well off when taking into account the togetherness value between spouses, which is a source of happiness in the relationship.

We calculate the indifference scales for 9 different types of households. We fix the individual utility at the level of marriage and assume flexible domestic productions, flexible inputs and 0 inputs from the spouses.

For most of the types, being single needs to obtain more than half of the household income

[3] The purple wave shown in u^t indicates the case that the mother has more decisive power, and the remaining line represents the case that father has more power.

	One child		Two children		Three children	
	Husband	Wife	Husband	Wife	Husband	Wife
First quartile's full income	0.7164	0.5377	0.7213	0.5595	0.7303	0.5688
Median full income	0.7200	0.5801	0.7223	0.5863	0.7233	0.5922
Third quartile's full income	0.7279	0.5981	0.7425	0.6009	0.8141	0.5947

Table 3: Indifference scales with fixed u^k, u^p and u^t

to achieve the same private utility level when in marriage. The counterfactual result shows that all singles need to achieve more than half of household income to maintain the same level of individual utilities and the argument acknowledges the importance of togetherness to spouses within the marriage.

5 Conclusion

We interpret the estimation results from three dimensions, which correspond to three mechanisms shown in the collective model.

First, individual preferences between two adult members. All three domestic goods have significant positive effects on both couples' individual utility. Women attach a higher marginal utility to children's welfare, chores and togetherness than men do. This indicates wives value togetherness more than husbands do.

Second, domestic production within households. As for the production technologies, we find that one extra time spent on children by the mother benefits more than one extra time spent on children by the father. An opposite pattern emerges for the household chores. For togetherness, one extra hour spent with spouses benefits togetherness more than 250 euro spent on joint activities with the spouse per week. As for the interaction between inputs, for u^k, the elasticity of substitution for an average household is 0.96, which indicates that the time spent by the father, the time spent by the mother and the expenditure on children are mutually complemented; for u^p, the average elasticity of substitution amounts to 1.02 so the inputs for chores present weak substitutability; for u^t, the average elasticity of substitution is 1.38 but with a fairly high standard deviation of 0.72 (the substitutability for togetherness inputs has a large heterogeneity across households). We further find the substitutability magnitude grows with the number of kids and presents an inverted-U shape correlation with the year of the relationship. The economic intuition is that households with more kids are easier to substitute expenditure and time for the production of togetherness joy. Married couples who have a short-term relationship are more likely to substitute time and consumption in producing togetherness joy, however after reaching a certain year, they become less likely to swap time and consumption on the joint activities.

Third, the bargaining effect between two adult members. The increasing nonlabor household income and increasing spousal age difference leads to decreasing bargaining power of males; while the higher relative wage leads to higher bargaining power of males.

Overall, this paper proposes a collective labor supply model with household production of togetherness outcomes to account for joint leisure and monetary expenditure that spouses spend on togetherness. The study suggests that time and consumption invested in togetherness are substitutes- spouses can substitute joint leisure with more upgraded services without harming their relationship. This implies spouses are sensitive to wage increases and it offers counter-evidence for spouses' working schedule synchronization. This study also reveals spouses' gender-related preferences and their bargaining power in household decision-making. We can link the findings to the analysis of gender differences or gaps rooted in households.

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A Identification Proof

After constructing the model, the critical step is to prove the identifiability of the model. Namely, the model is identifiable if the mapping from model structures to observed behaviour is one-to-one. In our case, the model is identifiable if at most one preference structure and bargaining position corresponds to the observed household behaviours including labour supply, private consumption, private leisure, childcare, chores and joint spousal activities.

We present the solution to the household optimization programme according to a two-stage allocation representation (Chiappori, 1988, 1992). At the first stage, the spouses agree on certain levels of domestic production (u^j; $j = k, p, t$) and the intrahousehold allocation of the residual nonlabor income (ρ^i; $i = 1, 2$). At the second stage, each member maximizes their utility by choosing private consumption and private leisure conditional on the domestic production outputs and personal budget agreed at the first stage.

Our proof breaks into three parts. In subsection A.1, we recover the production functions from observed input variables using proposition 1. In subsection A.2, we start with the second stage where we recover private preferences and thus the individual indirect utility functions up to a strictly increasing transformation on given levels of domestic output. In subsection A.3, we formulate the first stage and prove the transformation is unique, from which we show the complete generical identifiability of Pareto weight and individual preferences.

A.1 Identify production functions

A.1.1 Recover production functions from observables

We observe the input variables (c^k, h_k^1, h_k^2), (c^p, h_p^1, h_p^2) and (c^t, l^t) as functions of $(w_1, w_2, y, \mathbf{z}, \mathbf{s})$, from which we recover the unobserved production functions $u^k(c^k, h_k^1, h_k^2)$, $u^p(c^p, h_p^1, h_p^2)$ and $u^t(c^t, l^t)$ up to strictly increasing transformations.

We assume the production process is cost-minimizing, so we have for $u^j; j = k, p$

$$\frac{\frac{\partial u^j}{\partial h_j^1}}{\frac{\partial u^j}{\partial c^j}} = \phi_j^1(c^j, h_j^1, h_j^2; \mathbf{s}) = w^1; \frac{\frac{\partial u^j}{\partial h_j^2}}{\frac{\partial u^j}{\partial c^j}} = \phi_j^2(c^j, h_j^1, h_j^2; \mathbf{s}) = w^2; \tag{12}$$

and for $u^j; j = t$

$$\frac{\frac{\partial u^t}{\partial l^t}}{\frac{\partial u^t}{\partial c^t}} = \phi_t(c^j, h_j^1, h_j^2; \mathbf{s}) = w^1 + w^2 \tag{13}$$

All produced goods in this model are non-marketable with time as inputs. An identification issue associated with the empirical implementation of the time allocation model is that the values of the different time uses are not observable. A common approach to address this issue is to assume that each household member's possible time uses have a uniform price equal to the individual's market wage (Cherchye et al. (2015)) [4]. Based on this assumption, we set $\phi_j^1(c^j, h_j^1, h_j^2; s) = w^1$ and $\phi_j^2(c^j, h_j^1, h_j^2; s) = w^2$, and the value of joint time is the sum of individual wages as $\phi_t(c^j, h_j^1, h_j^2; s) = w^1 + w^2$.

The Eq.12 and Eq.13 form systems of partial differential equations (hereafter PDEs). In the systems, Eq.12, the first-order homogeneous linear partial differential equations outnumber the unknown coefficients and this leads to an overdetermined problem. The problem can be solved by Frobenius theorem (Palais, 2011). The main idea is to set some cross-derivates equal, so that some unknown effects are symmetric and thus the systems are still integrable and have a solution.

In this spirit, the PDEs ϕ_j^1 and ϕ_j^2 are integrable if and only if the ϕ_j^1 and ϕ_j^2 are compatible, that is, the Slutsky equations are satisfied:

$$\frac{\partial \phi_j^1(c^j, h_j^1, h_j^2; s)}{\partial h_j^2} + \frac{\partial \phi_j^1(c^j, h_j^1, h_j^2; s)}{\partial c_j}\phi_j^2 = \frac{\partial \phi_j^2(c^j, h_j^1, h_j^2; s)}{\partial h_j^1} + \frac{\partial \phi_j^2(c^j, h_j^1, h_j^2; s)}{\partial c_j}\phi_j^1.$$

In the production of $u^j(c^j, l^j); j = t$, the effect of time input is one-dimensional and thus

[4] The identification problem becomes more difficult in the case when the shadow prices are assumed non-uniform but remain unobservable. However, (Chiappori and Lewbel, 2015) provides an approach to deal with this issue.

ϕ_t is directly integrable by the Local Existence and Uniqueness theorem.

Overall, the production functions u^k, u^p and u^t can be recovered up to a strictly increasing transformation given the observable inputs as functions (w_1, w_2, y, z, s). Let $r_j(.); j = k, p, t$ be strictly increasing functions and we substitute u^j by $r_j(u^j); j = k, p, t$, then the individual utility function becomes $u^i(c^i, l^i, u^k, u^p, u^t) = u^i(c^i, l^i, r_k^{-1}(u^k), r_p^{-1}(u^p), r_t^{-1}(u^t))$ and leads to the same observable choices. Hence, the increasing transformation is a matter of normalization of the production function and it doesn't affect the optimal allocation.

Given that $u^j; j = k, p, t$ can be recovered inputs $(c^j, h_j^1, h_j^2; j = k, p)$ and $(c^j, l^j; j = t)$ and that the inputs are known functions of observable state variables (w^1, w^2, y, z, s), the production functions are observed and expressed as $u^k(w^1, w^2, y, z, s)$, $u^p(w^1, w^2, y, z, s)$ and $u^t(w^1, w^2, y, z, s)$.

A.1.2 Why do we need production shifters and distributional factors

We assume on two conditions about the production functions, a). production technologies are of constant-return-to-scale b). no shared inputs in different production processes. According to the standard demand analysis, the quantities of output can be expressed in functions of price indices and the full income. In the time allocation model, the expression leads to fundamental identification issue that models with different preferences are observationally equivalent and we cannot disentangle preferences from technologies (Cherchye et al., 2015). A common approach to deal with this issue is the usage of distribution factors and production shifters, see Blundell et al. (2005) and Cherchye et al. (2012).

The identification approach can be seen from Proposition 1, in which we have three continuously differentiable functions $u^j(w^1, w^2, y, z, s)$ where $j = (k, p, t)$, and we need at least two significant production shifters s^1, s^2 and one distributional factor z to ensure the identifiability.

Assuming that the Jacobian matrix $\begin{bmatrix} \frac{\partial u^k}{\partial z} & \frac{\partial u^k}{\partial s^1} & \frac{\partial u^k}{\partial s^2} \\ \frac{\partial u^p}{\partial z} & \frac{\partial u^p}{\partial s^1} & \frac{\partial u^p}{\partial s^2} \\ \frac{\partial u^t}{\partial z} & \frac{\partial u^t}{\partial s^1} & \frac{\partial u^t}{\partial s^2} \end{bmatrix}$ is invertible (nonsingular) in an appropriately defined subset of the domain of $u^j(w^1, w^2, y, z, s)$ where $j = (k, p, t)$, then we can express the distribution factor z and the production shifters s^1, s^2 as functions of $z = z(w^1, w^2, y, \overline{u}^k, \overline{u}^p, \overline{u}^t, z^-, s^-)$, $s^1 = s^1(w^1, w^2, y, \overline{u}^k, \overline{u}^p, \overline{u}^t, z^-, s^-)$, $s^2 = s^2(w^1, w^2, y, \overline{u}^k, \overline{u}^p, \overline{u}^t, z^-, s^-)$ by Implicit Function Theorem. We interpret these functions in that we need at least one distribution factor z and two production shifters s^1 and s^2 to keep domestic outputs constant while allowing variation in wages and nonlabor income. Alternatively, we can adopt three production shifters to keep the income variation. However, we cannot use two or more distribution factors because the distribution factors operate only through their impact on Pareto weight and their impact is one-dimensional (many distribution factors operate as if there was one distribution factor), so the Jacobian matrix may be singular by construction if we have two different distribution factors.

A.2 Identify preferences up to a strictly increasing transformation

After addressing the identification issue associated with domestic productions, we proceed to recover the conditional sharing rules by formulating a second-stage representation.

The conditional sharing rules are defined as the residual non-labour income after expenditures on the household production.

$$\rho^i(w^1, w^2, y, \mathbf{z}, \mathbf{s}) = w^i l^i(w^1, w^2, y, \mathbf{z}, \mathbf{s}) + c^i(w^1, w^2, y, \mathbf{z}, \mathbf{s}) - w^i. \tag{14}$$

Given the household budget constraint, we have

$$\begin{aligned}
&\rho^1(w^1, w^2, y, z, s) + \rho^2(w^1, w^2, y, z, s) = y - c^k(w^1, w^2, y, z, s) - w^1 h_k^1(w^1, w^2, y, z, s) \\
&- w^2 h_k^2(w^1, w^2, y, z, s) - c^p(w^1, w^2, y, z, s) - w^1 h_p^1(w^1, w^2, y, z, s) - w^2 h_p^2(w^1, w^2, y, z, s) \\
&- c^t - (w^1 + w^2) l_t(w^1, w^2, y, z, s).
\end{aligned} \tag{15}$$

We then formulate the second stage individual maximization program for $i = 1, 2$

$$\max_{c^i, l^i} u^i(c^i, l^i, \overline{u}^k, \overline{u}^p, \overline{u}^t) \tag{16}$$

subject to

$$c^i + w^i l^i = w^i + \rho^i.$$

Chiappori (1988, 1992) showed that we can recover the sharing rule up to an additive constant and the individual preferences up to a translation from the observation of private labor supply. Blundell et al. (2005) showed the individual sharing rule can be recovered up to an increasing constant depending on public consumption with the additional observation of public expenditure. Cherchye et al. (2012) extended the collective model with household production and showed that the sharing rules are exactly identified using detailed private consumption (c^1 and c^2). Our model follows the basic framework of Cherchye et al. (2012) and the individual preferences are recovered up to a strictly increasing transformation for given $\overline{u}^j, j = (k, p, t)$ from the observed private consumption and private leisure.

Summarizing, for any given $\overline{u}^j$ where $j = (k, p, t)$, we have two 3-tuples (u^1, u^2, ρ) and $(\hat{u}^1, \hat{u}^2, \hat{\rho})$ that generate the same solutions to program (16) for all income bundle (w^1, w^2, y) [5]. The relation between the two tuples can be shown as

$$\begin{aligned}
\hat{u}^1(c^1, l^1, \overline{u}^k, \overline{u}^p, \overline{u}^t) &= f^1(u^1(c^1, l^1, \overline{u}^k, \overline{u}^p, \overline{u}^t), \overline{u}^k, \overline{u}^p, \overline{u}^t) \\
\hat{u}^2(c^2, l^2, \overline{u}^k, \overline{u}^p, \overline{u}^t) &= f^2(u^2(c^2, l^2, \overline{u}^k, \overline{u}^p, \overline{u}^t), \overline{u}^k, \overline{u}^p, \overline{u}^t) \qquad (17) \\
\hat{\rho}(w^1, w^2, y) &= \rho(w^1, w^2, y),
\end{aligned}$$

where the functions $f^i, i = (1, 2)$ are increasing in respective $u^j, j = (k, p, t)$. Analogously, we obtain the same relations as of the indirect utility functions

$$\begin{aligned}
\hat{v}^1(w^1, \rho^1, \overline{u}^k, \overline{u}^p, \overline{u}^t) &= f^1(v^1(w^1, \rho^1, \overline{u}^k, \overline{u}^p, \overline{u}^t), \overline{u}^k, \overline{u}^p, \overline{u}^t) \\
\hat{v}^2(w^2, \rho^2, \overline{u}^k, \overline{u}^p, \overline{u}^t) &= f^2(v^2(w^2, \rho^2, \overline{u}^k, \overline{u}^p, \overline{u}^t), \overline{u}^k, \overline{u}^p, \overline{u}^t)
\end{aligned} \qquad (18)$$

so that the indirect utility functions are also identified up to a strictly increasing trans-

[5] For simplicity, here we denote $\rho = \rho^1$. Since the budget constraint is known, we easily derive ρ^2 after identifying ρ^1

formation that depends on the given level of production level. Suppose we know (v^1, v^2, ρ), then the $(\hat{v}^1, \hat{v}^2, \hat{\rho})$ will be known if the $f^i, i = (1, 2)$ is identified. Hence, our next step is to identify the f^i in the first stage allocation process.

A.3 The transformation is unique

In this subsection, we prove that the previous increasing transformation between two tuples is unique so that the transformation is identified. Furthermore, individual preference, production outputs, and the Pareto weight are all generically identified.

We assume the domestic production is cost-minimizing and define the expenditure functions of domestic goods $j = (k, p)$ as

$$e^j(u^j, w^1, w^2) = \min_{c^j, h^1_j, h^2_j} \{c^j + w^1 h^1_j + w^2 h^2_j | u^j(c^j, h^1_j, h^2_j; s^j) = u^j\} \tag{19}$$

and for $j = t$ as:

$$e^j(u^j, w^1, w^2) = \min_{c^j, l^j} \{c^j + (w^1 + w^2) l^j | u^j(c^j, l^j; s^j) = u^j\} \tag{20}$$

These expenditure functions give the minimal expenditures on the inputs needed to produce a fixed quantity of outputs. Since the production functions exhibit constant returns to scale as we assumed, the cost function can be expressed as

$$e^j(u^j, w^1, w^2) = g^j(w^1, w^2) u^j, \tag{21}$$

where g^j is a homogeneous price index for the domestic good j.

Then we proceed to formulate the first stage decision process by substituting the tuple $(\hat{v}^1, \hat{v}^2, \hat{\rho})$ into household utility maximization program

$$\max_{\rho^1, \rho^2, u^k, u^p, u^t} \lambda \hat{v}^1(w^1, \rho^1, u^k, u^p, u^t) + (1 - \lambda) \hat{v}^2(w^2, \rho^2, u^k, u^p, u^t) \tag{22}$$

subject to

$$\rho^1 + \rho^2 + g^k(w^1, w^2)u^k + g^p(w^1, w^2)u^p + g^t(w^1, w^2)u^t = y.$$

We derive the first-order conditions for the interior solution,

$$\begin{aligned}
\frac{\partial L}{\partial \rho^1} &= \lambda \frac{\partial \hat{v}^1}{\partial \rho^1} - \delta = 0 \\
\frac{\partial L}{\partial \rho^2} &= (1-\lambda) \frac{\partial \hat{v}^2}{\partial \rho^2} - \delta = 0 \\
\frac{\partial L}{\partial u^k} &= \lambda \frac{\partial \hat{v}^1(w^1, \rho^1, u^k, u^p, u^t)}{\partial u^k} + (1-\lambda) \frac{\partial \hat{v}^2(w^2, \rho^2, u^k, u^p, u^t)}{\partial u^k} - \delta g^k(w^1, w^2) = 0 \\
\frac{\partial L}{\partial u^p} &= \lambda \frac{\partial \hat{v}^1(w^1, \rho^1, u^k, u^p, u^t)}{\partial u^p} + (1-\lambda) \frac{\partial \hat{v}^2(w^2, \rho^2, u^k, u^p, u^t)}{\partial u^p} - \delta g^p(w^1, w^2) = 0 \\
\frac{\partial L}{\partial u^t} &= \lambda \frac{\partial \hat{v}^1(w^1, \rho^1, u^k, u^p, u^t)}{\partial u^t} + (1-\lambda) \frac{\partial \hat{v}^2(w^2, \rho^2, u^k, u^p, u^t)}{\partial u^t} - \delta g^t(w^1, w^2) = 0 \\
\frac{\partial L}{\partial \delta} &= y - \rho^1 - \rho^2 - g^k(w^1,w^2)u^k - g^p(w^1, w^2)u^p - g^t(w^1, w^2)u^t = 0.
\end{aligned} \tag{23}$$

where the δ is the shadow price for the budget constraint. We then rewrite the first-order conditions and obtain the following

$$\begin{aligned}
\lambda &= \frac{\frac{\partial \hat{v}^2(w^2,\rho^2,u^k,u^p,u^t)}{\partial \rho^2}}{\frac{\partial \hat{v}^1(w^1,\rho^1,u^k,u^p,u^t)}{\partial \rho^1} + \frac{\partial \hat{v}^2(w^2,\rho^2,u^k,u^p,u^t)}{\partial \rho^2}}, \\
g^j(w^1, w^2) &= \frac{\frac{\partial \hat{v}^1(w^1,\rho^2,u^k,u^p,u^t)}{\partial u^j}}{\frac{\partial \hat{v}^1(w^1,\rho^1,u^k,u^p,u^t)}{\partial \rho^1}} + \frac{\frac{\partial \hat{v}^2(w^1,\rho^2,u^k,u^p,u^t)}{\partial u^j}}{\frac{\partial \hat{v}^2(w^2,\rho^2,u^k,u^p,u^t)}{\partial \rho^2}}.
\end{aligned}$$

which are standard Lindahl conditions for the public goods. The left hand side is the sum of the marginal rate of substitution between the produced good j and the private commodity, and the right hand side is the price ratio for the two goods.

We compute the partial derivatives of $\hat{v}^j; i = (1, 2), j = (k, p, t)$ via equation A.2,

$$\begin{aligned}
\frac{\partial \hat{v}^i}{\partial \rho^i} &= \frac{\partial f^i}{\partial v^i} \frac{\partial v^i}{\partial \rho^i} \\
\frac{\partial \hat{v}^i}{\partial u^j} &= \frac{\partial f^i}{\partial v^i} \frac{\partial v^i}{\partial u^j} + \frac{\partial f^i}{\partial u^j}.
\end{aligned}$$

and substitute them in Eq.A.3, so that we have

$$\frac{1}{\frac{\partial v^1}{\partial \rho^1}}\frac{\frac{\partial f^1}{\partial u^j}}{\frac{\partial f^1}{\partial v^1}}+\frac{1}{\frac{\partial v^2}{\partial \rho^2}}\frac{\frac{\partial f^2}{\partial u^j}}{\frac{\partial f^2}{\partial v^2}}=g^j(w^1,w^2)-(\frac{\frac{\partial v^1}{\partial u^j}}{\frac{v^1}{\partial \rho^1}}+\frac{\frac{\partial v^2}{\partial u^j}}{\frac{v^2}{\partial \rho^2}}), \tag{24}$$

The price indices $g^j(w^1,w^2); j=k,p,t$ are known, and the indirect utility functions $v^i; i=1,2$ are identified up to strictly increasing transformation, so we can only identify the ratios $\frac{\partial f^i}{\partial u^j}/\frac{\partial f^i}{\partial v^i}$. And in the next step, we hope to prove the ratio is uniquely identified.

Define the ratio $\Theta^1_j(v^1,u^j)=\frac{\partial f^i}{\partial u^j}/\frac{\partial f^i}{\partial v^i}$ and rewrite the above equation, we obtain

$$\frac{1}{\frac{\partial v^1}{\partial \rho^1}}\Theta^1_j(v^1,u^j)+\frac{1}{\frac{\partial v^2}{\partial \rho^2}}\Theta^2_j(v^2,u^j)=g^j(w^1,w^2)-(\frac{\frac{\partial v^1}{\partial u^j}}{\frac{v^1}{\partial \rho^1}}+\frac{\frac{\partial v^2}{\partial u^j}}{\frac{v^2}{\partial \rho^2}}). \tag{25}$$

We assume by contradiction that there exists two solutions $\Theta^i_j(v^i,u^j)$ and $\Theta^{i'}_j(v^i,u^j)$ for Eq.A.3. The difference between two solutions are defined as $\phi^i_j(v^i,u^j)=\Theta^i_j(v^i,u^j)-\Theta^{i'}_j(v^i,u^j)$. By construct, the difference must satisfy the following differential equation

$$\frac{1}{\frac{\partial v^1}{\partial \rho^1}}\phi^1_j(v^1,u^j)+\frac{1}{\frac{\partial v^2}{\partial \rho^2}}\phi^2_j(v^2,u^j)=0.$$

or

$$\ln(|\phi^1_j(v^1,u^j)|)-\ln(|\phi^2_j(v^2,u^j)|)=\ln(\frac{\frac{\partial v^1}{\partial \rho^1}}{\frac{\partial v^2}{\partial \rho^2}}).$$

for $j=k,p,t$.

For general functions v^1, v^2 and ρ, the equation will not be satisfied unless $\phi^i_j(v^i,u^j)=0 (i=1,2; j=k,p,t)$, which indicates that the ratio solution to A.3 is generically unique.

Summarizing, the observation of $(c^i,l^i,h^1_k,h^i_p,l^t,c^k,c^p,c^t)$ as functions of (w^1,w^2,y,z,s) generically identifies the individual preferences, conditional sharing rules and the Pareto weights, so the model is identified.

B Consumption and Time Use variables composition

We apply our model to a sample of households drawn from the LISS panel (Longitudinal Internet Studies for the Social Sciences) covering 2009, 2010 and 2012. The LISS consists of questionnaires from different modules. We combine data from (1)Household box for background variables, (2)Core study 5 Family and Household, (3)Core study 6 Work and Schooling, (4)Assembled study 34 Time Use and Consumption. The last Module provides detailed consumption and time-use information within households, from which we obtain our key variables 'joint leisure time', the leisure time spent together with a partner, and 'consumption on joint activities with the spouse', which includes expenditure on eating at home, trips and holidays with a partner.

Public expenditure includes categories (1) expenditures on mortgages (rent and payment); (2) rent without expenditures on utilities; (3) utilities(heating, electricity, water, telephone, internet, etc. but not insurances);(4) transportation costs (public transport, gasoline, etc., but no insurances or purchase of transportation means); (5) insurances (house, car, health, etc.); (6) child care (kindergarten, after school care, guest parent, chores supervision, etc.); (7) alimony and financial support for children not (or no longer) living at home; (8) expenditures to service debt (but no mortgages); (9) trips and holidays with (part of) the family (airplane tickets, hotel, restaurant, etc.); (10) expenditures related to cleaning the house or gardening; (11) food and drinks consumed at home; and (12) other public expenditures not mentioned above. Recall that a follow-up question with respect to food and drinks consumed at home was added where all respondents had to indicate how much of these expenditures they personally consumed.

Chores expenditure (1) expenditures on mortgages (rent and payment): $bf09a066$; (2) rent without expenditures on utilities: $bf09a067$ (3) Utilities(heating, electricity, water, telephone, internet, etc. but not insurances);$bf09a068$ (4) transportation costs (public transport, gasoline, etc., but no insurance or purchase of transportation means): $bf09a069$, (5) insurances (house, car, health, etc.): $bf09a070$, (7) alimony and financial support for children not (or no longer) living at home: $bf09a072$ (8) expenditures to service debt (but no mortgages): $bf09a073$ (10) expenditures related to cleaning the house or gardening: $bf09a075$. (12) other public expenditures $bf09a077$

Childcare expenditure (6) child care (kindergarten, after-school care, guest parent, homework supervision, etc.): $bf09a071$ **Food and drinks expenditure: eating with children** $bf09a093$ Total expenditure per month for children living at home $bf09a114$ based on: food and drinks outside the house, cigarettes and other tobacco products, clothing, personal care products and services, medical care and health costs NOT covered by insurance, leisure time expenditure, (further)schooling, gifts and presents and other.

Public expenditure on joint activities (9) trips and holidays with (part of) the family (airplane tickets, hotel, restaurant, etc.): $bf09a074$ **monthly expenditure on eating at home by your partner:** $bf09a092$

Private consumption (1) food and drinks outside the home (restaurant, bar, company restaurant etc., but no expenditures consumed with (part of) the family: $bf09a095$; (2) cigarettes and other tobacco products: $bf09a096$; (3) clothing (clothing, shoes, jewellery, etc.): $bf09a097$; (4) personal care and services (hair care, body care, manicure, hair dresser etc., but no medical expenditures): $bf09a098$; (5) medical expenditures not covered by insurance (medicines, doctor, dentist, hospital, maternity grant, spectacles, hearing device, etc.): $bf09a099$; (6) leisure activities (film, theatre, hobbies, sports, photography, books, CDs, DVDs, expenditures related to travelling without the family, etc.): $bf09a100$; (7) schooling (courses, tuition fees, etc.) $bf09a101$; (8) gifts (to family members, friends, charity, etc.): $bf09a102$; (9) other private expenditures not mentioned above: $bf09a103$.

Monthly expenditure for your own personal consumption at home: eating by yourself $bf09a092$ and total expenditure per month for personal consumption based on above variables $bf09a104$.

Private time includes (1) paid work (excluding time spent on commuting); (2) commuting (for work or school); (3) domestic work (cleaning, dishwashing, cooking, shopping, gardening, do-it-yourself, etc., but no tasks related to caring for children or other persons); (4) personal care (washing, dressing, eating, visits to the hairdresser and doctor, etc.); (5) activities with children (washing, dressing, playing, reading, visiting the doctor, etc.); (6) helping parents (administrative tasks, washing, dressing, visiting the doctor, etc.); (7) helping other family members (administrative tasks, washing, dressing, visiting the doctor, etc.); (8) helping other persons who are not family members (administrative tasks, washing, dressing, visiting the doctor, etc.); (9) leisure activities (watching TV, reading, sports, hobbies, visiting friends or family, travelling, going out, etc.); (10) schooling (day or evening education, vocational training, language training, etc.); (11) administrative tasks related to own household; (12) sleeping and relaxing (sleeping, thinking, meditating, etc.); (13) other activities not mentioned above.

Chores (3) domestic work (cleaning, dish washing, cooking, shopping, gardening, do-it-yourself, etc., but no tasks related to caring for children or other persons): $bf09a009$, $bf09a010$ (11) administrative tasks related to own household: $bf09a025$, $bf09a026$

Childcare (5) activities with children (washing, dressing, playing, reading, visiting the doctor, etc.): $bf09a013$ $bf09a014$

Private leisure (4) personal care (washing, dressing, eating, visits to the hairdresser and doctor, etc.): $bf09a011$, $bf09a012$ (9) leisure activities (watching TV, reading, sports, hobbies, visiting friends or family, travelling, going out, etc.): $bf09a021$, $bf09a022$ (10) schooling (day or evening education, vocational training, language training, etc.): $bf09a023$, $bf09a024$ (12) sleeping and relaxing (sleeping, thinking, meditating, etc.): $bf09a027$, $bf09a028$.

Market work (1) paid work (excluding time spent on commuting): $bf09a005$, $bf09a006$; (2) commuting (for work or school): $bf09a007$, $bf09a008$

Joint leisure $bf09a064$, $bf09a065$